

Design Write-up

Architecture

The system replays one merged, time-ordered stream of ~52k quotes and 500 orders. A quote sharing a timestamp with an order is applied first, so every decision uses the most recent quote at or before “now”; the engine holds only the single current quote, never the tape, so lookahead is impossible by construction. All prices are integer cents. Five modules split the concerns: **data** (parsing), **book** (resting limits in price-time priority, a heap per side), **strategy** (all pricing and risk decisions), **engine** (execution, position, cash), **reporting**; paths and risk parameters live in `config.py`.

Two choices carry compliance. Every client fill passes through one chokepoint, `engine._fill()`, which checks NBBO bounds and the client’s limit before booking; a strategy bug crashes the replay instead of printing a violating fill. This covers the trap where the market moves through a resting limit: the order re-prices to the new touch or better, never its stale limit (`test_engine.py` pins both directions). Second, `validate.py` independently re-reads the raw CSVs and outputs, re-verifies every fill against the tape, reconstructs the position to zero, and recomputes the P&L.

Internalization strategy

The waterfall, for incoming orders and for resting orders re-evaluated on every tick, is cross, then internalize, then route. Crosses are riskless and improve both sides, so they win whenever limits overlap inside the NBBO; they are rare (1,600 shares) because marketable flow executes immediately rather than being held to fish for matches at the client’s expense.

Pricing. Principal fills price at the midpoint rounded a half-cent in the firm’s favor; at 2¢ this is the exact midpoint. The rule that an internalized client does strictly better than routing is self-imposed (the prompt requires only at-or-within NBBO); with whole-cent prices it makes 2¢ the minimum feasible spread for new risk, and the forgone 1¢ flow costs nothing since its half-cent edge equals the hedge cost. 2¢ is also the volume-maximizing choice; rerunning the day at 3¢:

	2¢ baseline	3¢ threshold
Internalized shares	243,300	97,400 (–60%)
Routed shares	232,200	378,100
Firm hedges	20 (21,300 sh)	10 (14,600 sh)
Realized P&L	\$14,396	\$13,499
Client price improvement	\$2,216	\$1,119 (–49%)

Raising the threshold forfeits the 2¢ bucket, the day’s largest: internalization drops 60% and client improvement halves while firm P&L barely moves. Two exceptions price differently: inventory-reducing flow at a 1¢ spread fills at the touch (client matches routing; firm exits without paying the spread), and the rebalancer fills resting inside-spread limits at the client’s own price, a cheaper exit than hedging.

Risk. A hard limit of ±10,000 shares is enforced by sizing: fills are truncated to remaining capacity, residual routed, so a breach cannot happen. A soft band of ±6,000 triggers immediate reduction, resting orders first, market second. From 15:55 the firm is reduce-only; at 16:00 it flattens (this day it was flat by 13:45 through client flow; the 17 leftover DAY orders expired, the only disposition that respects their limits). 10k is ~0.02% of AAPL’s ADV and ~1.6% of last-five-minutes volume, so the forced flatten is always liquid; the limit comes from the risk budget, not liquidity.

What is optimized, and the honest accounting. The objective is spread capture net of hedging costs, subject to never disadvantaging a client versus routing and to hard risk bounds. Attribution against the contemporaneous mid splits the day’s +\$14,396 exactly: +\$160 of execution edge (+\$545 from client fills, –

\$385 of hedge costs at 1.8¢/share) and +\$14,236 of inventory drift, \$11.4k of it in the 11:00 selloff hour when the book happened to be short. Midpoint pricing hands the client nearly the whole spread, so the profit is the position sitting on the right side of an engineered day: luck, which the band only bounds. Wide spreads cut both ways, most attractive to internalize exactly when the market trends and the exit costs half that same spread. Break-even: internalize only if expected capture from future offsetting flow exceeds the probability-weighted hedge cost plus expected adverse move.

Trade-offs accepted. No trend prediction (fitting a signal to one simulated day is curve-fitting; the band bounds the damage). No volatility estimate: one simulated day gives too few data points to measure it without forward-looking bias or extra assumptions, so thresholds and bands stay static. One refinement was tested and rejected: unwinding 500 shares whenever the spread is 1¢ and the book sits at the soft limit fired only 4 times all day, because pinned inventory and tight spreads anti-correlate — evidence the hedging problem is procyclical and must be solved before the book is pinned.

Assumptions

Routed orders and firm hedges fill fully at the touch regardless of displayed size (per the prompt). Same-millisecond quotes apply in file order; each fill records the exact quote it executed against. The close uses the session's last quote as the prevailing NBBO.

Production changes

1. **Symbols and feeds:** a feed handler per symbol pushing into per-symbol engine instances (the engine already consumes only an event stream); sequence numbers and staleness detection, since a stale NBBO silently invalidates “compliant” fills.
2. **Risk limits:** (1) capacity that shrinks with time remaining: $ADV \times (\text{remaining}/390) \times \text{participation}$ (5–10%) — at 5 minutes and 5% this gives ~32,000 shares, so 10,000 is conservative by 3x; (2) volatility-targeted bands with hysteresis (constant dollar risk; `position_after` pinned at $\pm 6,000$ shows the static band becoming a directional position in trends); (3) dollar limits, MOC for the flatten, a route-everything kill switch.
3. **Hedge execution:** choose between crossing the spread and posting inside it — cross only when a vol-based fair-value gap to the touch exceeds a threshold; work the rest passively.
4. **Pricing:** sub-penny improvement (makes 1¢ spreads internalizable); inventory-adjusted pricing that skews away from midpoint as the book fills; per-client markout tiers scaling the 2¢ threshold — treating all flow as equally benign is the biggest realism gap.
5. **Costs:** net exchange fees and market impact into internalize-vs-route.
6. **Data:** with tick/depth data, routed fills could walk the book instead of filling at the displayed touch.
7. **Order book for real message flow:** live flow is dominated by cancel/replace, where the heap book degrades to $O(n)$. Implemented as a swappable variant (`book_dict.py` / `engine_dict.py`): price-level dict + order-id index + sorted price ladder gives $O(1)$ cancel (hash to the node, pointer-surgery it out) with `on_cancel` semantics where a cancel racing a fill loses; a full-replay parity test proves fills, hedges, and cash identical to the heap engine. Data-structure choice follows the message mix: heaps for fill-only flow, the three-piece book once cancels arrive.
8. **Plumbing:** explicit OMS state machine with client execution reports, journaled fills/position for intraday restart, and `validate.py` run post-trade as an independent compliance process.

new to cancel replace order heapq is no longer optimal as retrieval takes $O(\log n)$